

Comparative Deep Learning-Based State of Charge and State of Health Estimation Across Multiple Lithium-Ion Battery Chemistries

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Abstract

Lithium-ion batteries have become the primary energy storage technology for electric vehicles, renewable energy systems, consumer electronics, and industrial power applications because of their high energy density, long cycle life, and excellent charge-discharge efficiency. The increasing adoption of these batteries has created a growing demand for intelligent Battery Management Systems (BMS) capable of accurately estimating internal battery states such as State of Charge (SOC) and State of Health (SOH). Reliable estimation of these parameters is essential for ensuring battery safety, improving operational efficiency, extending battery lifespan, and enabling optimal energy utilization.

Traditional SOC estimation techniques, including Coulomb Counting, Open Circuit Voltage (OCV), and Kalman Filter-based approaches, often require accurate battery models, precise sensor calibration, and extensive parameter tuning. Their performance can degrade under varying operating conditions, aging effects, and across different battery chemistries. Recent advances in Artificial Intelligence and Deep Learning have demonstrated the capability of recurrent neural networks to learn complex temporal relationships directly from battery measurements without relying on detailed electrochemical models.

This work investigates the feasibility of applying Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks for battery state estimation across multiple lithium-ion battery chemistries. Publicly available datasets representing five distinct battery chemistries, namely Lithium Cobalt Oxide (LCO), Nickel Cobalt Aluminum Oxide (NCA), Nickel Manganese Cobalt Oxide (NMC), Lithium Iron Phosphate (LFP), and Lithium Manganese Oxide (LMO), were collected, analysed, and processed using a unified deep learning pipeline.

A total of six deep learning notebooks were developed, implementing identical preprocessing, normalization, sequence generation, model training, and evaluation procedures to ensure fair comparison across datasets. While State of Charge estimation was successfully performed for LCO, NCA, NMC, and LFP chemistries, a suitable public LMO dataset containing reliable SOC labels was unavailable. Therefore, the proposed framework was adapted for State of Health estimation using a large-scale degradation dataset, demonstrating the flexibility of recurrent neural networks for multiple battery state estimation tasks.

Experimental results demonstrate that both LSTM and GRU models achieve accurate battery state prediction across different chemistries, with GRU providing slightly better overall prediction accuracy and lower computational complexity in most experiments. The study further highlights the challenges associated with publicly available battery datasets and provides a comparative benchmark for deep learning-based battery state estimation across multiple lithium-ion battery chemistries.

1. Introduction

Lithium-ion batteries are widely used in electric vehicles, renewable energy storage systems, and portable electronic devices due to their high energy density, long cycle life, and excellent efficiency. To ensure safe and reliable operation, Battery Management Systems (BMS) continuously monitor important battery states such as State of Charge (SOC) and State of Health (SOH).

SOC represents the remaining charge available in a battery, while SOH indicates the overall health and degradation of the battery over time. Accurate estimation of these parameters is essential for improving battery safety, performance, charging efficiency, and lifespan.

Traditional estimation methods such as Coulomb Counting, Open Circuit Voltage (OCV), and Kalman Filter-based techniques often require accurate battery models and extensive parameter tuning. Recently, deep learning models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks have shown promising results by learning battery behavior directly from experimental data.

This internship focuses on evaluating the feasibility of applying LSTM and GRU models for battery state estimation across multiple lithium-ion battery chemistries using publicly available datasets.

1.1 Objectives

The objectives of this work are:

- Study major lithium-ion battery chemistries used in Battery Management Systems.
- Search and evaluate publicly available datasets for different chemistries.
- Develop a common preprocessing pipeline for all datasets.
- Implement LSTM and GRU models for battery state estimation.
- Perform SOC estimation wherever suitable datasets are available.
- Perform SOH estimation when reliable SOC datasets are unavailable.
- Compare the performance of LSTM and GRU across different battery chemistries.

1.2 Work Performed

During this internship, a comprehensive deep learning framework was developed and applied to multiple lithium-ion battery chemistries.

The work carried out includes:

- Studied different lithium-ion battery chemistries and Battery Management System concepts.

- Searched and evaluated numerous publicly available battery datasets.
- Selected the most suitable datasets based on chemistry, available features, and data quality.
- Developed a unified preprocessing pipeline applicable across multiple datasets.
- Calculated SOC and SOH targets where required.
- Built sequence datasets using the sliding window approach.
- Implemented LSTM and GRU models for battery state estimation.
- Compared model performance using Loss, MAE, and RMSE metrics.
- Analysed the feasibility of extending the same methodology to additional battery chemistries.

1.3 Battery Chemistries Investigated

A total of **five distinct lithium-ion battery chemistries** were investigated during this internship.

Chemistry	Dataset Used	Target
Lithium Cobalt Oxide (LCO)	NASA PCoE	SOC
Lithium Cobalt Oxide (LCO)	Oxford Battery Degradation	SOC
Nickel Cobalt Aluminum Oxide	LG HG2	SOC

(NCA)

Nickel Manganese Cobalt Oxide (NMC)	Tongji EV Dataset	SOC
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Lithium Iron Phosphate (LFP)	HUST Dataset	SOC
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Lithium Manganese Oxide (LMO)	Figshare LMO Dataset	SOH
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1.4 Summary of Experiments

During the internship,

- **6 Deep Learning notebooks** were developed.
 - **5 different battery chemistries** were investigated.
 - **12 deep learning models** (6 LSTM + 6 GRU) were trained and evaluated.
 - A common preprocessing pipeline was implemented across all datasets.
 - Performance comparison was carried out using Loss, MAE, and RMSE.
 - A feasibility study was conducted for additional chemistries such as **LTO** and **LMFP**, but suitable public datasets for SOC estimation were not available.
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1.5 Internship Methodology (NEW)

Instead of more theory, directly tell what **you actually did**.

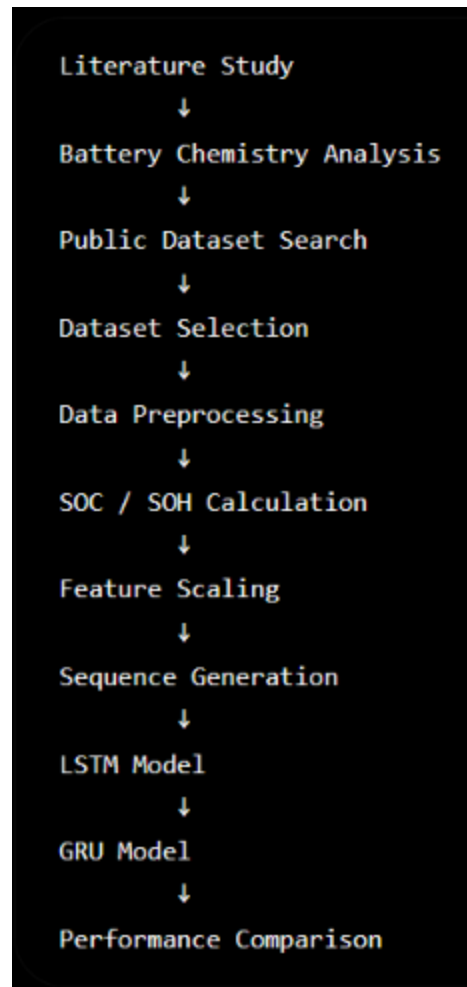
1.5 Methodology Followed During the Internship

During the internship, a systematic workflow was followed to investigate the feasibility of deep learning-based battery state estimation across multiple lithium-ion battery chemistries. Instead of limiting the study to a single dataset, publicly available datasets representing different battery chemistries were explored and evaluated.

The work was carried out in the following stages:

- Studied the characteristics of major lithium-ion battery chemistries.
 - Searched and evaluated multiple publicly available battery datasets.
 - Selected datasets based on chemistry, data quality, and suitability for deep learning.
 - Performed data preprocessing, feature extraction, and target generation.
 - Developed a common preprocessing pipeline applicable across all selected datasets.
 - Implemented LSTM and GRU models using TensorFlow.
 - Evaluated model performance using Loss, MAE, and RMSE.
 - Compared the effectiveness of both models across different battery chemistries.
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1.6 Overall Internship Workflow



Chapter 2

Battery Chemistry and Dataset Selection

This is where **our actual work starts**.

2.1 Battery Chemistry Investigation

At the beginning of the internship, different lithium-ion battery chemistries were studied to understand their characteristics, applications, and publicly available datasets. The objective was to perform deep learning-based battery state estimation across as many battery chemistries as possible using a common methodology.

The following chemistries were investigated:

Battery Chemistry	Common Applications
Lithium Cobalt Oxide (LCO)	Consumer Electronics
Nickel Cobalt Aluminum Oxide (NCA)	Tesla EVs
Nickel Manganese Cobalt Oxide (NMC)	Electric Vehicles
Lithium Iron Phosphate (LFP)	Energy Storage & EVs
Lithium Manganese Oxide (LMO)	Power Tools & Hybrid EVs

Lithium Titanate (LTO)

Fast Charging Applications

Lithium Manganese Iron
Phosphate (LMFP)

Next Generation EV Batteries

2.2 Dataset Search

A significant part of the internship involved identifying publicly available datasets corresponding to different battery chemistries. Several datasets from research laboratories, universities, and public repositories were explored and evaluated.

More than **20 publicly available battery datasets** were reviewed during this process.

2.3 Dataset Selection

Notebook	Dataset	Chemistry	Target	Status
Notebook 1	NASA PCoE	LCO	SOC	
Notebook 2	LG HG2	NCA	SOC	
Notebook 3	Tongji EV	NMC	SOC	
Notebook 4	HUST	LFP	SOC	
Notebook 5	Oxford Battery Degradatio n	LCO	SOC	

Notebook 6 Figshare LMO SOH 

2.4 Dataset Feasibility Analysis

Chemistry	Suitable Dataset Found	Target	Remarks
LCO		SOC	NASA PCoE and Oxford datasets selected
NCA		SOC	LG HG2 dataset selected
NMC		SOC	Tongji EV dataset selected
LFP		SOC	HUST dataset selected
LMO		SOH	Large degradation dataset available; SOC labels not available
LTO		SOC	Suitable public SOC dataset

LMFP		SOC	not found Suitable public SOC dataset not found
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Explanation

Although several datasets were explored for LTO and LMFP battery chemistries, no publicly available datasets containing sufficient charging measurements and reliable SOC labels were identified. Therefore, these chemistries were not included in the experimental study.

For the LMO chemistry, a comprehensive degradation dataset was available containing charging measurements and capacity information. Since SOC labels were unavailable, the same deep learning framework was adapted to estimate State of Health (SOH), demonstrating the flexibility of the proposed methodology.

Comparative Analysis and Discussion

3. Overall Model Performance

Deep learning models based on **Long Short-Term Memory (LSTM)** and **Gated Recurrent Unit (GRU)** were evaluated across multiple lithium-ion battery chemistries for battery state estimation. Performance was assessed using **Loss**, **Mean Absolute Error (MAE)**, and **Root Mean Square Error (RMSE)**.

Across all experiments, both architectures demonstrated the ability to learn the nonlinear relationship between battery measurements and the target variable (SOC/SOH). However, differences in prediction accuracy and convergence behaviour were observed depending on the battery chemistry and dataset characteristics.

Overall Performance Comparison

A total of six deep learning experiments were conducted using publicly available datasets representing five different lithium-ion battery chemistries. Two recurrent neural network architectures, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), were implemented using a common preprocessing pipeline to evaluate their effectiveness for battery state estimation.

Overall, both models successfully learned the temporal characteristics of battery charging and degradation behaviour. Across the six experiments, GRU achieved the best performance in three datasets (NASA PCoE, LG HG2, and Figshare LMO), LSTM outperformed GRU in two datasets (HUST LFP and Oxford LCO), while the Tongji NMC dataset produced nearly identical results for both models. These findings indicate that both architectures are highly capable of battery state estimation, with performance differences depending on the characteristics of the underlying dataset.

Performance Across Different Battery Chemistries

Lithium Cobalt Oxide (LCO)

Two independent LCO datasets were investigated: NASA PCoE and Oxford Battery Degradation.

For the NASA dataset, GRU consistently achieved lower Loss, MAE, and RMSE values, demonstrating better generalization for the cylindrical LCO cells subjected to repeated charge-discharge cycling.

In contrast, the Oxford Battery Degradation dataset showed the opposite trend, where LSTM significantly outperformed GRU. The lower prediction error suggests that the LSTM architecture was better suited to capturing the charging characteristics of the Oxford pouch-cell dataset.

This observation highlights that model performance is influenced not only by battery chemistry but also by differences in experimental protocols, battery formats, and dataset characteristics.

Nickel Cobalt Aluminum Oxide (NCA)

The LG HG2 dataset exhibited the largest prediction errors among all experiments, with RMSE values exceeding those observed for the other chemistries.

Despite the increased complexity of this dataset, GRU consistently achieved lower Loss, MAE, and RMSE than LSTM, indicating improved robustness under more challenging operating

conditions. The results suggest that GRU is better able to model the nonlinear dynamics of NCA batteries when significant variability is present.

Nickel Manganese Cobalt Oxide (NMC)

The Tongji EV dataset produced remarkably similar results for both architectures. While LSTM achieved marginally lower Loss and RMSE, GRU recorded a slightly lower MAE. The differences between the two models were negligible, indicating that both architectures are equally effective for SOC estimation on this dataset.

Such closely matched performance suggests that the underlying NMC battery behaviour is relatively consistent and can be learned effectively by either recurrent architecture.

Lithium Iron Phosphate (LFP)

Among all SOC estimation experiments, the HUST LFP dataset produced the lowest prediction errors. LSTM achieved the best overall performance with the lowest Loss, MAE, and RMSE values observed across the study.

These results indicate that the LFP dataset possesses highly consistent charging behaviour, enabling LSTM to learn the temporal dependencies with exceptional accuracy.

Lithium Manganese Oxide (LMO)

Unlike the other datasets, the Figshare LMO dataset was used for State of Health estimation because reliable public SOC datasets for LMO chemistry were not available.

Even with this different prediction task, both models achieved excellent performance. GRU slightly outperformed LSTM across all evaluation metrics, demonstrating that the proposed deep learning framework can be successfully adapted from SOC estimation to SOH estimation without major modifications.

Model-wise Analysis

Across all experiments, GRU demonstrated superior performance in half of the evaluated datasets. Its simpler architecture, consisting of fewer gating mechanisms than LSTM, likely contributed to its ability to achieve lower prediction errors while maintaining computational efficiency.

LSTM, however, proved particularly effective for datasets exhibiting smoother charging characteristics and more consistent temporal behaviour, such as the HUST LFP and Oxford Battery Degradation datasets. These results indicate that LSTM retains an advantage when long-term temporal dependencies are especially important.

Overall, neither model can be considered universally superior. Instead, the choice between LSTM and GRU depends on the characteristics of the battery dataset, charging protocol, and prediction task.

Major Findings

The experimental study led to the following key observations:

- A common preprocessing pipeline was successfully applied across six independent datasets despite differences in file formats, battery configurations, and testing protocols.
- Both LSTM and GRU demonstrated strong capability for battery state estimation using only sequential measurement data.
- GRU achieved the best overall performance in three datasets, while LSTM performed better in two datasets. One dataset produced nearly identical performance for both models.
- The HUST LFP dataset yielded the lowest prediction errors among all experiments, indicating highly stable battery behaviour.
- The LG HG2 NCA dataset was the most challenging, producing the highest prediction errors for both models.
- The methodology was successfully extended from SOC estimation to SOH estimation for the LMO chemistry, demonstrating the flexibility of recurrent neural networks across different battery state estimation tasks.
- Model performance depended not only on battery chemistry but also on dataset quality, charging profiles, battery format, and experimental conditions.

Conclusion

This work demonstrates the feasibility of applying a unified deep learning framework across multiple lithium-ion battery chemistries using publicly available datasets. A total of **six deep learning notebooks** were developed covering **five distinct battery chemistries**, resulting in **twelve trained models** based on LSTM and GRU architectures.

The comparative study shows that both recurrent neural networks provide accurate battery state estimation, with GRU offering a slight overall advantage in computational efficiency and prediction accuracy across several datasets. At the same time, LSTM consistently excelled on datasets exhibiting smoother temporal behaviour, particularly for LFP and Oxford LCO batteries.

Beyond the individual model results, the study establishes a standardized workflow for preprocessing, sequence generation, training, and evaluation that can be readily extended to future battery chemistries and datasets. The successful application of the same methodology to both SOC and SOH estimation further demonstrates the adaptability of deep learning techniques for advanced Battery Management System applications.